

① What are the different protocols you observe at the following layers of the protocol stack?

- A) Application Layer
- B) Transport Layer
- C) Network Layer

Soln

① Application Layer

- HTTP (Hypertext Transfer Protocol)
- DNS (Domain Name System)
- TLS (Transport Layer Security)
- OSCP (Online Certificate Status Protocol)
- QUIC

② Transport Layer

- TCP (Transmission Control Protocol)
- UDP (User Datagram Protocol)

③ Network Layer

- IPv4 (Internet Protocol Version 4)
- IPv6 (Internet Protocol Version 6)
- ICMP (Internet Control Message Protocol)
- ICMPv6 (Internet Control Message Protocol Version 6)

Q) 2) What is the total amount of data being received for the following two cases?

A) When you access "http://neverssl.com/"

B) When you access "https://www.cornell.edu"

Soln

(A) When accessing (http://neverssl.com/)

→ 1498 (bytes) = 1.46 (Kilo bytes)

(B) When accessing (~~http~~ https://www.cornell.edu)

→ 180000 (bytes) = 180 (Kilo bytes)

Q) 3) Answer the following when you access the Site (<http://www.neverssl.com/>)

Soln

(A) How many HTTP GET request do you observe, list down the GET requests

→ Total HTTP GET request : 1

→ GET request: GET / HTTP/1.1

(B) What version of HTTP the server is running.

→ HTTP version: HTTP/1.1

(C) For each of the HTTP GET request, find out:

(i) The total number of TCP Segment being received.

→ 6 TCP Segment

(ii) Total number ~~number~~ amount of data being received in the corresponding HTTP response message.

→ 1900 bytes.

⑤ Identify TCP three way handshake used to establish the connection with the <https://www.neverssl.com>.
Specific SYN, SYN-Ack, ACK packets and write the sequence numbers for each of the three accesses.

→ TCP (Three way handshake)

* Packet	Description	Sequence Number	Acknowledged Number
9	SYN	0	—
93	SYN-ACK	0	1
94	ACK	1	1